

PRAVEEN DEDIGAMAGE

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RESEARCH INTERESTS

Large-scale brain network modelling; Neuron–astrocyte interactions and synaptic plasticity; Multiscale neural simulation (NEST, Brian2); High-performance and exascale computing; Spiking neural networks; Biomedical signal processing (EEG/EMG); Neuromorphic computing

EDUCATION

Master's Programme in Biomedical Sciences and Engineering 08/2022 – 12/2024

Tampere University, Tampere, Finland — GPA: 4.38/5.0

- Major: Biomedical Micro- and Nanodevices
- Thesis: A spiking neural network-based pipeline for motor imagery EEG classification (grade: very good)

Bachelor of the Science of Engineering Honours 11/2016 – 12/2021

University of Ruhuna, Galle, Sri Lanka — GPA: 3.48/4.0 (2nd Class Upper Division)

- Field of Specialization: Electrical and Information Engineering
- Gold medal for best final-year undergraduate research project (Dept. of EIE, 2021)

RESEARCH & PROFESSIONAL EXPERIENCE

Research Associate 12/2025 – 06/2026

Semiconductor Components and Microelectronic Systems Group, Technical University of Berlin

- Exploring memristive devices for event-based gas sensing

Teaching and Research Assistantship 12/2024 – 12/2025

Computational Neuroscience (CNS) Group, Tampere University

- Implemented a fully neuromorphic-compatible inference pipeline for motor imagery EEG classification
- Designed spiking-based spatial filters for real-time signal inferencing
- Conduct exercise sessions of the Introduction to Neuroinformatic course

Analog IC Design Traineeship 08/2024 – 09/2024

Neuromorphic Cognitive Systems Group, Institute of Neuroinformatics (INI), University of Zurich & ETH Zurich

- Designed an analog front end coupled with an Asynchronous Delta Modulator for tactile sensing

Research Assistantship 05/2023 – 08/2023

Computational Neuroscience (CNS) Group, Tampere University

- Abstract modelling of Pre-Pulse Inhibition (PPI) phenomena using the Brian2 simulator
- Compartmental modelling of GABA-B receptor effects on cortical neurons

Prototype Engineering Internship 06/2021 – 09/2021

Ceylon Business Appliances (Private) Limited, Colombo, Sri Lanka

- Designed an active dry electrode for EEG data acquisition systems
- Developed firmware and worked with biomedical data acquisition and analysis (ECG, HR, GSR)

RELEVANT TECHNICAL SKILLS

Simulation & Modelling: NEST simulator; Brian2; NEURON; The Virtual Brain (TVB); SnnTorch; COMSOL
HPC & Programming: C/C++; CUDA; Python; Open MPI; MATLAB; Git; LaTeX;

PUBLICATIONS

D.R.P. Chamal, M.-L. Linne, T. Mäki-Marttunen, “Fully Analog Compatible Spiking Neural Network Pipeline for Motor Imagery EEG Trial Classification,” (In preparation, to be submitted).

D.R.P. Chamal, M.-L. Linne, T. Mäki-Marttunen, “A Spiking Neural Network Based Pipeline for Motor Imagery EEG Classification,” LFTY Day 2025, February 13–14, 2025, Tampere University.

D.R.P. Chamal, M.I.M. Fernando, W.P.M.W. Kulathunga, K.D. Pathirana, N.W. Prins, “Design of a Novel Current Controlling Module for Functional Electrical Stimulation (FES) System,” 2021 6th ICITR, pp. 1–5, doi: 10.1109/ICITR54349.2021.9657272.

RELEVANT PROJECTS

- Preliminary study of implementing a memristive device for event-based gas sensing (2026)
- Spatially enhance casual encoding method for motor imagery EEG signals (2025)
- SNN-based pipeline for motor imagery EEG classification (2025)
- Analog front end + Asynchronous Delta Modulator for tactile sensing (2024)
- EFI-based closed-loop Deep Brain Stimulation circuit design (2024)
- Compartmental modelling of GABA-B effects on cortical neurons (2023)
- Network-level abstract neuron model for simulating PPI phenomena (2023)
- FES-based Machine Aided Hand Therapy for Post-Stroke Rehabilitation + BCI (2019–2021)
- Real-Time EMG Classification on FPGA (2019)

CONFERENCES & WORKSHOPS

- 10th EBRAINS Baltic-Nordic Summer School on Neuroscience, 28–31.05.2024
- Telluride Neuromorphic Cognition Engineering Workshop (Remote), 29.06–19.07.2025
- LFT-Day – Finnish Society for Medical Physics and Medical Engineering, 13–14.02.2025
- ICNCE 2024 – International Conference on Neuromorphic Computing and Engineering, 03–06.06.2024
- IEEE WE SYP Congress, EPFL University, Switzerland, 15–17.09.2023
- 11th Winter School “Ethics of Neuroscience and AI”, Bernstein Center, Berlin (Remote), 2022
- Virtual BCI & Neurotechnology Spring School 2021, organized by g.tec medical engineering GmbH

SCHOLARSHIPS & AWARDS

- PROFI6 Health Data Science Working Grant (4 months), Tampere University, 2025
- Invited traineeship at the Institute of Neuroinformatics (INI), ETH Zurich/UZH, 2024
- Finland Scholarship 2022 – 100% tuition coverage + €5,000 relocation grant
- Gold Medal for Best Final-Year Research Project, Dept. of EIE, University of Ruhuna, 2021

REFERENCES

Marja-Leena Linne, Ph.D., Docent

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Faculty of Medicine and Health Technology, Tampere University
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